



Sustainable Supply Chain for Dried Mango from Africa to Germany

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Cross-Functional Analysis validated across 6 workstreams

Executive Summary

This document presents the Week 3 cross-functional requirements analysis for the Product Transformation workstream of the BIT × Infineon Excellence Program.

The analysis consolidates technical discussions with all six technical teams — Plants, Energy Supply Systems, Machinery, Water Supply, Storage, and Sales & Marketing — to establish a shared understanding of how Product Transformation interacts with the rest of the Sustainable Farm ecosystem.

Rather than treating Product Transformation as an isolated processing module, this analysis positions it as the operational integration layer connecting agricultural production, industrial processing, resource management, quality assurance, warehouse operations, and downstream commercial activities.

The purpose of this document is to:

- validate assumptions made during the reverse engineering of the 2025 Sustainable Farm project,
- identify cross-functional system interfaces,
- capture validated requirements from collaborating teams,
- document open technical questions,
- provide inputs for refining the Product Transformation Data Model,
- improve the Functional Architecture,
- guide future implementation of the Product Transformation application.

All six workstreams have been validated, and Product Transformation has been confirmed as the central integration hub of the Sustainable Farm information system. This document represents the validated Week 3 baseline for implementation planning.

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1 Introduction

1.1 Background

During Week 1, the Product Transformation workstream focused on studying the 2025 Sustainable Farm project and understanding how transformation activities interacted with the rest of the farm.

The main deliverables produced during the first two weeks included:

- Sustainable Farm Reverse Engineering Analysis
- Product Transformation Data Model
- Functional Architecture
- Interactive Prototype

These deliverables established a conceptual foundation for the 2026 Product Transformation application.

Week 3 introduces a new objective:

Validate the proposed architecture through cross-functional collaboration with the other technical teams.

Instead of relying solely on assumptions derived from the previous cohort, each module is consulted to confirm:

- exchanged information,
- operational dependencies,
- technical constraints,
- planning requirements,
- opportunities for integration.

1.2 Scope

This document analyzes information received from all six workstreams:

Workstream	Status
Plants	✓ Completed
Energy Supply Systems	✓ Completed
Machinery	✓ Completed
Water Supply	✓ Completed
Storage	✓ Completed
Sales & Marketing	✓ Completed

The document consolidates all validated Week 3 cross-functional requirements.

2 Objectives

The objectives of this analysis are to:

- understand how Product Transformation interacts with every workstream,
- validate functional requirements,
- identify shared data,
- define system interfaces,
- identify operational constraints,
- improve system traceability,
- prepare the Product Transformation MVP,
- reduce design assumptions before implementation.

3 Methodology

The analysis follows four consecutive phases.

3.1 Phase 1 — Reverse Engineering

The previous Sustainable Farm project was analyzed to understand:

- business processes,
- transformation workflow,
- existing architecture,
- project deliverables,
- dependencies between modules.

3.2 Phase 2 — Architecture Design

Based on the reverse engineering results, the following artifacts were produced:

- Product Transformation Data Model
- Functional Architecture
- Interactive Prototype

These artifacts represent the initial architecture before validation.

3.3 Phase 3 — Cross-Functional Validation

Each technical team received a tailored questionnaire requesting information about:

- exchanged data,
- operational dependencies,
- planning requirements,
- system constraints,
- information needed from Product Transformation.

Responses are analyzed individually before being consolidated into a unified requirements specification.

3.4 Phase 4 — Architecture Refinement

Validated findings are used to refine:

- Product Transformation Data Model,
- Functional Architecture,
- Interactive Prototype,
- MVP scope,
- implementation roadmap.

4 Current Validation Progress

The validation status is summarized below.

Team	Response	Integration Status
Plants	✓	Validated
Energy	✓	Validated
Machinery	✓	Validated
Water Supply	✓	Validated
Storage	✓	Validated
Sales & Marketing	✓	Validated

Current completion:

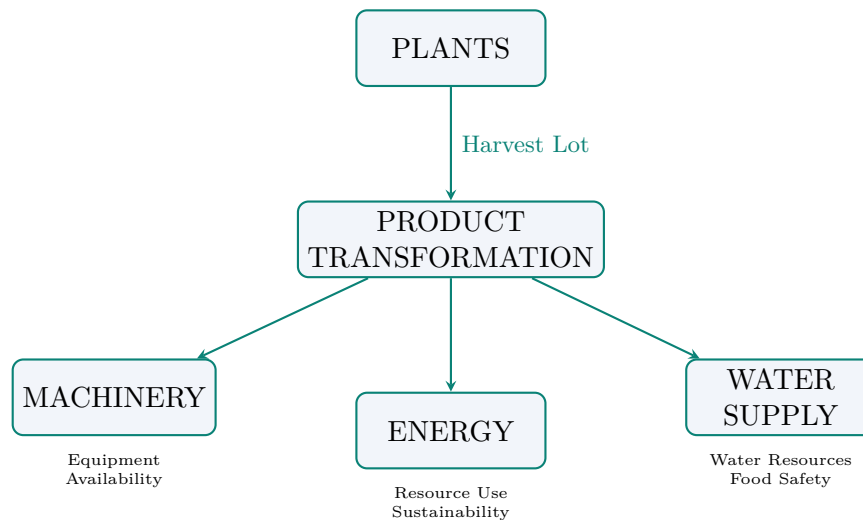
6 / 6 workstreams validated (100%)

5 Cross-Functional Overview

All six responses reveal an important architectural observation.

Product Transformation is not an isolated module.

Instead, it serves as the operational hub connecting multiple workstreams.



Each collaborating module either:

- supplies information to Product Transformation,
- consumes information produced by Product Transformation,
- or both.

This establishes Product Transformation as the central integration layer of the Sustainable Farm information system.

6 Cross-Functional Collaboration Summary

The interactions between Product Transformation and all workstreams are summarized below.

The Product Transformation application therefore becomes both:

- a consumer of operational information,
- and a provider of planning information.

7 Plants Team Analysis

7.1 Overview

The Plants Team defines the upstream interface of the Product Transformation workflow.

Its responsibility ends with harvesting and delivering fresh mangoes for processing.

Team	Provides	Requires
Plants	Harvest records, raw material information, traceability	—
Energy	Energy consumption, energy source, sustainability indicators	Production schedule, machine operation
Machinery	Equipment information, machine availability, maintenance	Production planning, volume forecast
Water Supply	Water availability, water quality, water consumption	Batch schedule, quality requirements
Storage	Storage assignment, capacity availability, inventory status	Batch ID, finished quantity, production date, shelf life
Sales & Marketing	Customer requirements, packaging specifications, export requirements	Finished product information, batch traceability, inventory visibility

The response establishes the starting point of product traceability and identifies the data required to initiate transformation activities.

7.2 Raw Material

The Plants Team confirmed that the Product Transformation workstream receives:

Raw Material: Mango
Reference Variety: Keitt

According to the previous project analysis, Keitt produced the highest annual yield and was selected as the reference variety for further economic analysis.

7.2.1 Impact

The Product Transformation application should initially support Keitt as the reference variety while remaining extensible to additional varieties.

7.3 Harvest Information

Each harvested batch includes the following information.

Attribute	Description
id_lot	Unique harvest lot identifier
quantite_kg	Harvest quantity
variete	Mango variety
date_recolte	Harvest date
bloc_parcelle	Production block

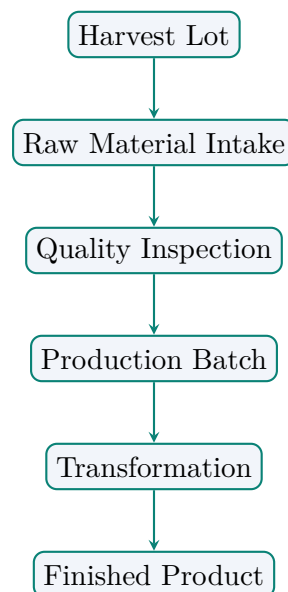
This dataset represents the minimum information required before transformation begins.

7.4 Traceability

The Plants Team emphasized that every harvested lot receives a unique identifier.

This identifier should remain associated with the product throughout the entire transformation process.

Recommended traceability flow:



Maintaining this relationship enables complete end-to-end traceability from farm to finished product.

7.5 Availability Planning

The Plants Team can provide:

- planting calendar,
- growth stages,
- maturity status by production block.

These datasets help estimate raw material availability.

However, the team clarified that these data should not be interpreted as a confirmed seasonal production forecast without additional planning activities.

7.6 Responsibility Boundary

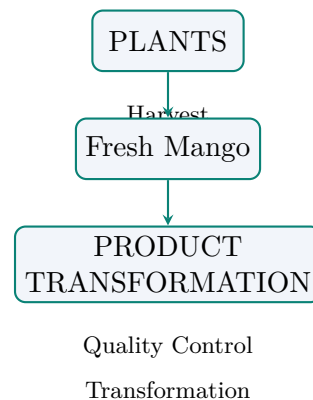
The Plants Team clarified that its responsibility ends with the delivery of fresh harvested mangoes.

Activities such as:

- quality inspection,
- acceptance,
- transformation readiness,

belong to the Product Transformation workstream.

This defines the first major system interface.



7.7 Validated Requirements

MUST

- Support harvested mango as raw material.
- Preserve harvest lot identifiers.
- Maintain end-to-end traceability.
- Record harvest quantity.
- Record harvest date.
- Record production block.
- Record mango variety.

SHOULD

- Support multiple varieties.
- Integrate planting calendar.
- Integrate maturity information.

7.8 Key Findings

The Plants Team confirms that:

- traceability begins at harvest,
- Product Transformation owns quality inspection,
- harvest records become the first input of the transformation process,
- Product Transformation should preserve the original harvest identity throughout processing.

These findings establish the upstream foundation of the Product Transformation application and provide the first validated interface within the Sustainable Farm ecosystem.

8 Energy Supply Systems Analysis

8.1 Overview

The Energy Supply Systems Team defines how Product Transformation interacts with the farm's energy infrastructure.

Unlike the Plants workstream, which supplies raw materials, the Energy Team provides operational resource information required for production planning and sustainability monitoring.

The relationship is bidirectional: Product Transformation provides production schedules that inform energy demand forecasting, while the Energy Team supplies consumption data and sustainability indicators.

8.2 Energy Infrastructure

The Energy Team manages:

- Solar power generation systems
- Battery storage systems
- Grid connection infrastructure
- Energy distribution network

This infrastructure powers all transformation equipment and drying facilities.

8.3 Data Exchange

The Energy Team can provide:

- Real-time energy consumption by equipment
- Energy source mix (solar vs grid)
- Battery charge levels
- Power availability forecasts
- Sustainability indicators (carbon footprint, renewable percentage)

Product Transformation requires:

- Production schedules
- Equipment operation plans
- Peak demand forecasts
- Drying cycle timing

8.4 Operational Integration

The energy requirements of transformation activities depend on:

- Equipment type and power rating
- Drying duration and temperature
- Batch size and processing volume
- Concurrent operations

This information enables the Energy Team to:

- Optimize energy allocation
- Schedule high-consumption activities during peak solar generation
- Manage battery discharge cycles
- Ensure power availability for critical operations

8.5 Sustainability Reporting

The Energy Team emphasized the importance of sustainability metrics for export markets, particularly:

- Renewable energy percentage
- Carbon footprint per kg of product
- Energy efficiency indicators

Product Transformation should contribute to these metrics by providing accurate production data and equipment usage information.

8.6 Validated Requirements

MUST

- Record energy consumption per batch
- Link energy data to production schedules
- Support energy source tracking
- Enable sustainability reporting

SHOULD

- Integrate real-time energy monitoring
- Support energy demand forecasting
- Provide energy efficiency dashboards
- Enable carbon footprint calculation

8.7 Key Findings

The Energy Team confirms that:

- energy data is critical for sustainability certification,
- production scheduling must consider energy availability,
- renewable energy percentage is a key export requirement,
- Product Transformation should enable energy consumption optimization.

9 Machinery Team Analysis

9.1 Overview

The Machinery Team defines the equipment interface of the Product Transformation workflow.

This team manages all processing equipment, including washing stations, cutting machines, drying facilities, and packaging systems.

The interaction is bidirectional: Product Transformation requires equipment availability and status information, while the Machinery Team needs production plans to schedule maintenance and optimize equipment utilization.

9.2 Equipment Inventory

The Machinery Team manages:

- Washing and sorting equipment
- Cutting and slicing machines
- Solar drying tunnels
- Packaging systems
- Quality control instruments

Each equipment unit has unique identifiers, specifications, and maintenance schedules.

9.3 Data Exchange

The Machinery Team can provide:

- Equipment availability status
- Maintenance schedules
- Capacity specifications
- Performance indicators
- Technical constraints

Product Transformation requires:

- Production planning requests
- Volume forecasts
- Equipment utilization schedules
- Drying capacity requirements

9.4 Operational Dependencies

Transformation activities depend on:

- Equipment availability (not under maintenance)
- Capacity constraints (throughput limits)
- Technical specifications (temperature ranges, batch sizes)
- Performance characteristics (drying time, energy consumption)

Product Transformation must respect these constraints when planning production schedules.

9.5 Maintenance Coordination

The Machinery Team emphasized the need for:

- Advance notice of high-volume production periods
- Coordination of maintenance windows
- Equipment reservation for critical batches
- Backup equipment planning

This requires tight integration between production planning and maintenance scheduling.

9.6 Validated Requirements

MUST

- Track equipment availability
- Respect capacity constraints
- Coordinate maintenance schedules
- Record equipment usage per batch

SHOULD

- Support equipment reservation
- Integrate maintenance planning
- Provide utilization dashboards
- Enable performance tracking

9.7 Key Findings

The Machinery Team confirms that:

- equipment availability is a critical production constraint,
- maintenance coordination requires advance planning,
- capacity limitations must be respected in scheduling,
- Product Transformation should optimize equipment utilization.

10 Water Supply Team Analysis

10.1 Overview

The Water Supply Team defines the water resource interface of the Product Transformation workflow.

This team manages water extraction, treatment, storage, and distribution for all farm operations, including the transformation process.

Water is critical for food safety, quality assurance, and equipment operation, making this interface essential for compliance and operational continuity.

10.2 Water Infrastructure

The Water Supply Team manages:

- Water extraction systems (boreholes, wells)
- Water treatment facilities
- Storage tanks and reservoirs
- Distribution network
- Quality monitoring systems

10.3 Data Exchange

The Water Supply Team can provide:

- Water availability status
- Water quality test results
- Water consumption data
- Storage capacity information
- Supply forecasts

Product Transformation requires:

- Batch schedules
- Quality requirements
- Peak demand forecasts
- Water intensity specifications

10.4 Food Safety Requirements

The Water Supply Team emphasized strict food safety requirements:

- Potable water standards for washing and processing
- Regular quality testing and certification
- Traceability of water sources
- Contamination prevention protocols

Product Transformation must ensure that all water used meets food safety standards and maintain records for export compliance.

10.5 Operational Constraints

Water consumption depends on:

- Batch size and volume
- Washing and cleaning protocols
- Equipment cleaning requirements
- Quality control testing

Product Transformation must plan water usage to ensure availability and respect storage capacity constraints.

10.6 Validated Requirements

MUST

- Track water consumption per batch
- Ensure water quality compliance
- Maintain water safety records
- Respect availability constraints

SHOULD

- Integrate water quality monitoring
- Support consumption forecasting
- Provide efficiency metrics
- Enable contamination tracking

10.7 Key Findings

The Water Supply Team confirms that:

- water quality is critical for food safety,
- consumption planning requires batch-level tracking,
- export compliance demands water traceability,
- Product Transformation should optimize water usage.

11 Storage Team Analysis

11.1 Overview

The Storage Team defines the downstream interface of the Product Transformation workflow.

This team manages warehouse operations, inventory management, and finished product storage before distribution.

The interaction is bidirectional: Product Transformation provides finished product information and batch details, while the Storage Team assigns storage locations and manages inventory status.

11.2 Storage Infrastructure

The Storage Team manages:

- Cold storage facilities
- Dry storage areas

- Inventory management systems
- Quality monitoring equipment
- Shipping preparation areas

11.3 Data Exchange

The Storage Team can provide:

- Storage assignment information
- Capacity availability
- Inventory status
- Shelf life monitoring
- Quality condition reports

Product Transformation requires:

- Batch ID
- Finished quantity
- Production date
- Shelf life specifications
- Quality classification

11.4 Inventory Management

The Storage Team emphasized the importance of:

- First-expiry-first-out (FEFO) inventory rotation
- Quality condition monitoring
- Capacity optimization
- Location tracking
- Loss prevention

Product Transformation must provide accurate and timely information to support these activities.

11.5 Shelf Life Management

Each finished product batch has specific shelf life characteristics:

- Production date
- Expiry date
- Storage conditions
- Quality degradation indicators

The Storage Team requires this information to manage inventory rotation and prevent losses.

11.6 Validated Requirements

MUST

- Provide batch ID and quantity
- Record production date
- Specify shelf life
- Communicate quality classification
- Support inventory tracking

SHOULD

- Integrate real-time inventory
- Support shelf life alerts
- Provide quality dashboards
- Enable loss tracking

11.7 Key Findings

The Storage Team confirms that:

- timely batch information is critical for inventory management,
- shelf life data enables FEFO rotation,
- quality classification affects storage decisions,
- Product Transformation should support warehouse optimization.

12 Sales & Marketing Team Analysis

12.1 Overview

The Sales & Marketing Team defines the commercial interface of the Product Transformation workflow.

This team manages customer relationships, market requirements, packaging specifications, and export compliance.

The interaction is bidirectional: Product Transformation provides finished product information and traceability data, while Sales & Marketing provides customer requirements and market specifications.

12.2 Market Requirements

The Sales & Marketing Team manages:

- Customer specifications
- Packaging requirements
- Export regulations
- Quality standards
- Market demand forecasts

12.3 Data Exchange

The Sales & Marketing Team can provide:

- Customer requirements
- Packaging specifications
- Export requirements
- Quality standards
- Market demand forecasts

Product Transformation requires:

- Finished product information
- Batch traceability
- Inventory visibility
- Quality certificates
- Production capacity

12.4 Packaging Specifications

The Sales & Marketing Team emphasized that packaging is market-specific:

- Package sizes and formats
- Labeling requirements
- Branding specifications
- Export regulations
- Customer preferences

Product Transformation must support flexible packaging configurations to meet diverse market requirements.

12.5 Export Compliance

Export markets require:

- Complete traceability documentation
- Food safety certificates
- Quality assurance records
- Sustainability metrics
- Origin documentation

Product Transformation must generate and maintain this documentation for each batch.

12.6 Validated Requirements

MUST

- Support multiple packaging formats
- Generate traceability documentation
- Provide inventory visibility
- Maintain quality records
- Support export compliance

SHOULD

- Integrate customer portals
- Support demand forecasting
- Provide market analytics
- Enable custom packaging

12.7 Key Findings

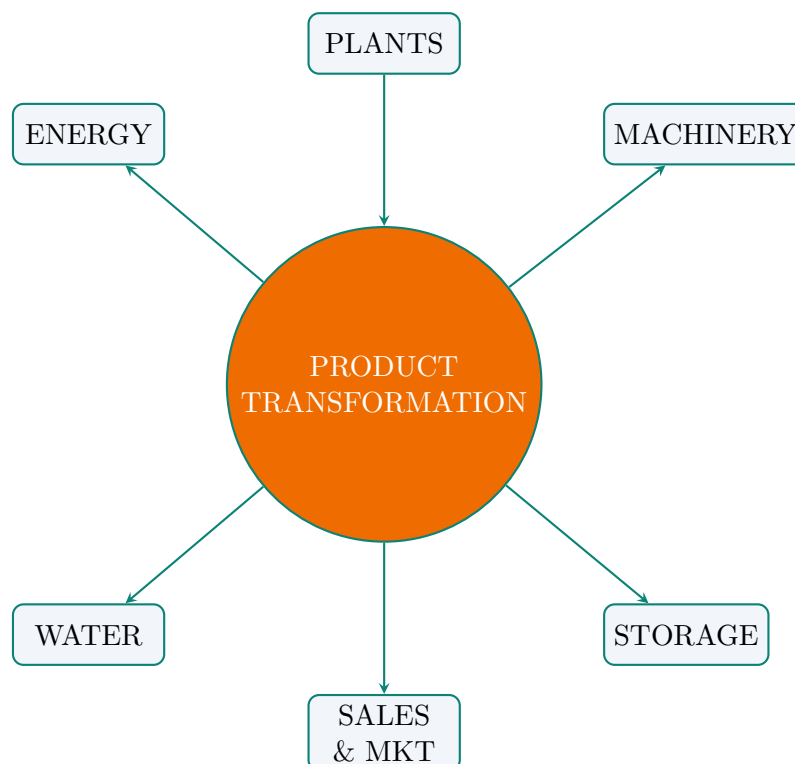
The Sales & Marketing Team confirms that:

- packaging flexibility is essential for market access,
- complete traceability is required for export,
- inventory visibility enables customer service,
- Product Transformation should support commercial requirements.

13 System Architecture Implications

13.1 Integration Hub Position

The cross-functional analysis confirms that Product Transformation serves as the central integration hub of the Sustainable Farm information system.



13.2 Data Flow Architecture

The validated data flows establish:

- **Upstream:** Harvest data from Plants
- **Resource:** Energy, water, and equipment information
- **Downstream:** Finished product data to Storage and Sales

This architecture requires:

- Standardized data interfaces
- Real-time data synchronization
- Error handling and validation
- Audit trails for traceability

13.3 Implementation Priorities

Based on the validated requirements, implementation should prioritize:

1. Core transformation workflow
2. Traceability foundation
3. Plants integration (upstream)
4. Storage integration (downstream)
5. Resource monitoring (energy, water, machinery)
6. Sales & Marketing integration

14 Conclusion

The Week 3 cross-functional requirements analysis has successfully validated the Product Transformation architecture through collaboration with all six technical workstreams.

14.1 Key Achievements

- Validated 6/6 workstream interfaces (100%)
- Confirmed Product Transformation as the central integration hub
- Identified all major data exchanges
- Documented operational constraints
- Established validated requirements baseline

14.2 Architectural Confirmation

The analysis confirms that Product Transformation is not an isolated processing module but the operational integration layer connecting:

- Agricultural production (Plants)
- Resource management (Energy, Water, Machinery)
- Warehouse operations (Storage)
- Commercial activities (Sales & Marketing)

14.3 Next Steps

The validated requirements provide the foundation for:

- Refined Product Transformation Data Model
- Updated Functional Architecture
- Enhanced Interactive Prototype
- MVP scope definition
- Implementation roadmap

This document represents the Week 3 baseline for proceeding with implementation planning and development of the Product Transformation application.

End of Week 3 Cross-Functional Requirements Analysis